Homework 5

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1 Inverse Comb Filter

$$y_t = x_t - R^L x_{t-L},\ R = 0.999,\ L = 6$$

1.1 Transfer Function $\mathcal{H}(z)$

$$\begin{aligned} Y &= X(1 - R^L z^{-L}) \\ \mathcal{H}(z) &= 1 - 0.999^6 z^{-6} \end{aligned}$$

1.2 Frequency Response $H(\omega)$

$$\begin{aligned} \mathcal{H}(e^{i\omega}) &= 1 - R^L e^{-iL\omega} \\ H(\omega) &= 1 - 0.999^6 e^{-i6\omega} \end{aligned}$$

1.3 Magnitude Response $|H(\omega)|$

$$\begin{aligned} H(\omega) &= e^{-iL\omega}(e^{iL\omega} - R^L) \\ |H(\omega)| &= |e^{iL\omega} - R^L| \\ |H(\omega)| &= |\cos(L\omega) + i\sin(L\omega) - R^L| \\ |H(\omega)| &= \sqrt{(\cos(L\omega) - R^L)^2 + \sin^2(L\omega)} \\ |H(\omega)| &= \sqrt{\cos^2(L\omega) - 2R^L\cos(L\omega) + R^{2L} + \sin^2(L\omega)} \\ |H(\omega)| &= \sqrt{1 - 2R^L\cos(L\omega) + R^{2L}} \\ |H(\omega)| &= \sqrt{(1 + 0.999^{12}) - 2(0.999)^6\cos(6\omega)} \end{aligned}$$

1.4 Maxima and Minima of $|H(\omega)|$

To find the maxima, one must find all values of ω where $\cos(6\omega) = -1$. Therefore, the following maxima can be found:

$$\omega = \frac{k\pi}{L},\ k = 2n + 1,\ n \in \mathbb{Z}$$

Similar to the maxima, minima are all values of ω where $\cos(6\omega) = 1$. Therefore, the following minima can be found:

$$\omega = \frac{k\pi}{L},\ k = 2n,\ n \in \mathbb{Z}$$

To convert these to sampling rate, use identity $\omega = 2\pi\frac{f}{f_s}$. By substitution, the following is derived:

$$f_{\max} = \frac{kf_s}{2L},\ k = 2n+1$$
$$f_{\min} = \frac{kf_s}{2L},\ k = 2n$$

Where $n \in \mathbb{Z}$
This results in the following points:

$$\max(|H(\omega)|_{dB}) \approx \{(\frac{f_s}{12}, 5.99), (\frac{3f_s}{12}, 5.99), (\frac{5f_s}{12}, 5.99)\}$$
$$\min(|H(\omega)|_{dB}) \approx \{(0, -44.5), (\frac{2f_s}{12}, -44.5), (\frac{4f_s}{12}, -44.5), (\frac{6f_s}{12}, -44.5)\}$$

1.5 Sketch of $|H(\omega)|_{dB}$

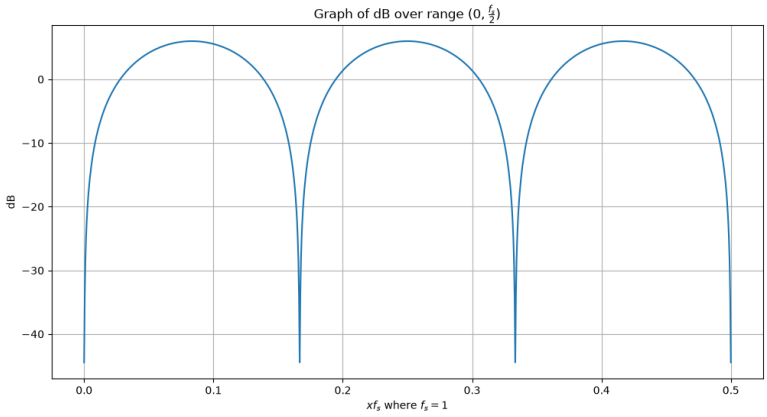


Figure 1: Magnitude Response Graph in dB

1.6 Phase response

$$
\begin{aligned}
H(\omega) &= 1 - R^L e^{-iL\omega} \\
H(\omega) &= 1 - R^L(\cos(-L\omega) + i\sin(-L\omega)) \\
H(\omega) &= (1 - R^L\cos(-L\omega)) + R^L i\sin(-L\omega) \\
\theta(\omega) &= \arctan(\frac{-R^L\sin(L\omega)}{(1 - R^L\cos(L\omega))}) \\
\theta(\omega) &= \arctan(\frac{-0.999^6\sin(6\omega)}{(1 - 0.999^6\cos(6\omega))})
\end{aligned}
$$

2 Designing a Reson Filter

One must first convert to radians per sample:

$$
\begin{aligned}
f_s &= 44100\text{Hz} \\
B_{\text{rad}} &= 2\pi\frac{B_{\text{Hz}}}{f_s}\ \text{rad/sample} \\
\psi_{\text{rad}} &= 2\pi\frac{\psi_{\text{Hz}}}{f_s}\ \text{rad/sample}
\end{aligned}
$$

Radians will be the default unit for the rest of the problem.

2.1 With $B = 2\phi,\ \cos\phi = 2 - \frac{1}{2}(R + \frac{1}{R})$

2.1.1 Deriving the pole radius formula

$$
\begin{aligned}
\phi &= \frac{10\pi}{f_s} \\
\cos\phi &= 2 - \frac{1}{2}(R + \frac{1}{R})
\end{aligned}
$$

Following the steps from class notes, one will obtain:

$$
\begin{aligned}
0 &= R^2 - 2(2 - \cos\phi)R + 1 \\
R &= \frac{2(2 - \cos(\phi)) \pm \sqrt{[-2(2 - \cos(\phi))]^2 - 4}}{2}
\end{aligned}
$$

2.1.2 Evaluating with $B = 10\text{Hz},\ \psi = 1000\text{Hz}$

Following the steps results in the filter:

$$\begin{aligned} y_t \approx & 0.00020216605481176457x_t \\ & +1.9783246236903536y_{t-1} \\ & -0.998576255974041y_{t-2} \end{aligned}$$

The impulse response is the following:

$$\begin{aligned} y_0 &= 0.00020216605481176457 \\ y_1 &= 0.00039995008430844753 \\ y_2 &= 0.00058935287793546 \end{aligned}$$

2.1.3 Evaluating with $B = 20\text{Hz},\ \psi = 5000\text{Hz}$

Following the steps results in the filter:

$$\begin{aligned} y_t \approx & 0.0018598988543063856x_t \\ & +1.511461125281618y_{t-1} \\ & -0.9971545393551153y_{t-2} \end{aligned}$$

The impulse response is the following:

$$\begin{aligned} y_0 &= 0.0018598988543063856 \\ y_1 &= 0.0028111648152399217 \\ y_2 &= 0.00239435974968163 3 \end{aligned}$$

2.2 With approximation $B = 2(1 - R)$

2.2.1 Evaluating with $B = 10\text{Hz},\ \psi = 1000\text{Hz}$

Following the steps results in the filter:

$$\begin{aligned} y_t = & 0.0002022380575210094x_t \\ & +1.978324121049828y_{t-1} \\ & -0.9985757489111912y_{t-2} \end{aligned}$$

The impulse response is the following:

$$\begin{aligned} y_0 &= 0.0002022380575210094 \\ y_1 &= 0.00040009242738807546 \\ y_2 &= 0.00058956248000382 \end{aligned}$$

2.2.2 Evaluating with $B = 20\text{Hz},\ \psi = 5000\text{Hz}$

Following the steps results in the filter:

$$\begin{aligned} y_t &= 0.0018612234956407x_t \\ &+1.5114595871848688y_{t-1} \\ &-0.9971525127908784y_{t-2} \end{aligned}$$

The impulse response is the following:

$$\begin{aligned} y_0 &= 0.0018612234956407 \\ y_1 &= 0.002813164096379871 \\ y_2 &= 0.0023960601582540676 \end{aligned}$$

2.3 Comment

They are suprisingly closer than I expected, it makes sense it's so widely used.

3 Filter Reverse Engineering

$$y_t = \sqrt{2}(0.99^t)\sin(\frac{\pi}{4}(t+1))$$

This filter can be matched with the formula:

$$y_t = R^t[\frac{\sin(\theta(t+1))}{\sin\theta}]$$

From there we can get the following:

$$\begin{aligned} R &= 0.99 \\ \theta &= \frac{\pi}{4} \end{aligned}$$

4 $R \approx e^{-\frac{B}{2}}$ Approximation Derivation

Suppose $R = 1 - \epsilon$, then one can represent $\cos\phi$ as:

$$\cos\phi = 2 - \frac{1}{2}(1 - \epsilon + (1-\epsilon)^{-1})$$

Then from the power series of $(1-\epsilon)^{-1}$:

$$\cos\phi = 2 - \frac{1}{2}(1 - \epsilon + 1 + \epsilon + \epsilon^2 + \dots)$$

After simplifying a bit:

$$\cos\phi = 1 - \frac{1}{2}\epsilon^2 + \dots$$

Using the expansion of $\cos\phi$:

$$\cos\phi = 1 - \frac{1}{2}\phi^2 + \dots$$

From there, we can find the approximation $\phi \approx \epsilon \approx \frac{1}{2}B$. Substituting accordingly gives:

$$R \approx 1 - \frac{1}{2}B$$

Looking at the expansion of decaying exponential:

$$e^{-x} \approx 1 - x$$

We can then deduce our approximation from there:

$$R \approx e^{-B/2}$$

5 Discrepancies of pole angles and peak frequencies